

NANDHAKUMAR M

SENIOR .NET FULL-STACK DEVELOPER | TECHNICAL LEAD

+91 80721-27298 | nandhakumarmurugesan45@gmail.com | Namakkal, Tamil Nadu, India

LinkedIn: linkedin.com/in/nandhakumar-m | GitHub: github.com/Nandhakumarmnk | Portfolio: nandhakumarmnk.github.io/portfolio

PROFESSIONAL SUMMARY

Senior .NET Full-Stack Developer and Technical Lead with **5 years of experience** architecting and delivering enterprise-scale supply chain, procurement, and business intelligence platforms using ASP.NET Core, ASP.NET MVC, C#, .NET Core Web API, and SQL Server. Led a four-phase enterprise security remediation program that resolved **500+ OWASP Top 10 vulnerabilities** across a live production platform, and built SAP synchronization services from scratch that eliminated manual cross-team reconciliation. Proven in owning zero-downtime production releases on IIS with no SLA breaches, tuning high-volume T-SQL workloads to sub-second response times, and mentoring developers on secure coding and .NET best practices. Skilled in AI-assisted development using Claude Code and ChatGPT to accelerate feature delivery and strengthen code review quality.

TECHNICAL SKILLS

Backend: C#, .NET Core, ASP.NET Core, ASP.NET MVC 5, .NET Core Web API, REST APIs, Entity Framework 6, LINQ, Dependency Injection, Swagger / OpenAPI, API Versioning

Frontend: JavaScript, TypeScript, jQuery, Bootstrap, HTML5, CSS3, Razor Views, Responsive Web Design

Database: Microsoft SQL Server, T-SQL, Stored Procedures, Query Optimization, Indexing Strategies, Execution Plan Analysis, Backup & Recovery

Security: OWASP Top 10, JWT Authentication, Role-Based Access Control (RBAC), XSS Prevention, CSRF Protection, CSP Headers, Input Validation, Parameterized Queries, DOMPurify, Secure File Upload

DevOps & Deployment: Git, TFS, Azure DevOps, IIS Administration, Hangfire, Zero-Downtime Deployments, Postman, Production Support

Integrations: SAP Integration, RESTful Services, SMTP / Notification Services, Excel Bulk Data Pipelines, Background Job Scheduling

Reporting & BI: DevExpress, Crystal Reports, KPI Dashboards, Drill-Down Data Visualization, SQL Aggregation & Analytics

Leadership & Practices: Technical Mentoring, Code Review, Secure Coding Standards, Requirement Analysis, Release Management, Stakeholder Coordination

AI Tools: Claude Code, ChatGPT, Prompt Engineering, AI-Assisted Code Review & Development

PROFESSIONAL EXPERIENCE

Senior .NET Full-Stack Developer & Technical Lead

Apr 2023 – Present

Spark IT Tech, Tiruppur, Tamil Nadu

- Lead the technical direction of the enterprise supply chain and procurement platform, owning architecture decisions, release planning, and production stability across multiple concurrent modules.
- Directed a four-phase enterprise security remediation program that identified and resolved **500+ vulnerabilities** spanning XSS, SQL Injection, CSRF, and unsafe file upload paths on a live production system.
- Established secure coding standards for the development team and conduct recurring code reviews and security audits to enforce them across all new development.
- Mentor junior developers on .NET best practices, defensive coding, SQL performance tuning, and AI-assisted development workflows.
- Introduced AI-assisted development practices using Claude Code and ChatGPT across the engineering team, improving feature turnaround and code review depth.
- Own end-to-end IIS deployments, server configuration, and zero-downtime production releases for procurement and finance systems with **no SLA breaches**.
- Coordinate with business stakeholders to translate procurement, finance, and inventory requirements into scalable technical designs and delivery timelines.

.NET Full-Stack Developer

Aug 2021 – Mar 2023

Spark IT Tech, Tiruppur, Tamil Nadu

- Built and shipped core procurement and inventory management modules supporting high-volume daily transactions on the enterprise SCM platform.
- Developed RESTful .NET Core APIs with JWT authentication, API versioning, and Swagger documentation consumed by internal web and integration clients.
- Optimized complex T-SQL stored procedures and indexing strategies to achieve **sub-second response times** under peak transactional load.
- Designed Excel bulk upload modules with multi-layer validation, tooltip-driven templates, and SAP synchronization override logic, cutting operational onboarding time.
- Configured Hangfire recurring jobs with retry policies, queue management, and monitoring dashboards to move long-running workflows off the request path.

KEY PROJECTS

Supply Chain & Procurement Management System — Enterprise Internal Platform

Technologies: C#, ASP.NET MVC, .NET Core, Web API, Entity Framework, SQL Server, Hangfire, JWT, IIS, Azure DevOps, Git

- Architected and delivered a multi-module platform covering procurement, inventory, and supply chain operations for high-volume daily transactions.
- Designed the RESTful API layer with JWT authentication, versioning, and Swagger documentation for internal and integration consumers.
- Enforced 150-character validation rules consistently across Razor views, APIs, service layers, and Excel workflows, shipped with **zero production defects**.
- Tuned stored procedures, query plans, and indexes to sustain sub-second response times as transaction volume scaled.

Enterprise Security Remediation Initiative — Lead Developer, Security | 4-Phase Program

Technologies: ASP.NET Core, C#, SQL Server, DOMPurify, CSP Headers, OWASP Top 10, JWT, RBAC, Parameterized Queries

- Led a four-phase remediation program across a live enterprise SCM platform, resolving **500+ vulnerabilities** without disrupting business operations.
- Applied DOMPurify sanitization, parameterized queries, and Content Security Policy headers to close XSS and SQL Injection attack surfaces.
- Implemented RBAC, XSS prevention filters, CSRF protection, and centralized input validation across every application entry point.
- Ran code reviews and security audits and codified the resulting secure coding standards for ongoing team use.

SAP Synchronization & Integration Services

Technologies: C#, .NET Core, Web API, Hangfire, SAP Integration, SQL Server, SMTP

- Built SAP synchronization services from scratch using Hangfire for near real-time procurement and finance data exchange.
- Eliminated manual cross-team reconciliation by automating bidirectional master and transactional data flows.
- Added retry policies, failure alerting, and queue monitoring so integration failures surface and self-recover without manual intervention.

BI Reporting & KPI Dashboard — Oct 2023 – Jan 2024

Technologies: C#, ASP.NET MVC, SQL Server, DevExpress, Crystal Reports, JavaScript, jQuery, Bootstrap

- Engineered real-time BI dashboards and KPI reporting modules using DevExpress controls and advanced SQL aggregation queries.
- Built Crystal Reports integrations for both scheduled and on-demand executive reporting.
- Designed drill-down data visualization layers giving stakeholders actionable insight into procurement and supply chain KPIs.
- Used Claude Code and ChatGPT to compress feature development and code review cycles.

Excel Bulk Upload & Data Validation Framework

Technologies: C#, ASP.NET MVC, .NET Core, SQL Server, SAP Integration, JavaScript

- Designed a reusable bulk upload framework with template generation, tooltip comments, and row-level validation feedback.
- Implemented SAP synchronization override logic to reconcile bulk-loaded records against master data safely.
- Reduced operational onboarding effort for high-volume master data entry across procurement modules.

KEY ACHIEVEMENTS

- Resolved **500+ security vulnerabilities** across a live enterprise SCM platform, aligning the application with OWASP Top 10 standards.
- Built SAP integration workflows from scratch, removing recurring cross-team reconciliation effort and enabling near real-time synchronization.
- Delivered measurable SQL performance gains on high-volume transactional modules through stored procedure and indexing optimization.
- Led zero-downtime production deployments for procurement and finance systems with **no SLA breaches**.
- Introduced AI-assisted development workflows within the engineering team, improving delivery speed and code quality.

EDUCATION

Bachelor of Engineering (B.E.) – Electrical & Electronics Engineering

2016 – 2020

Karpagam College of Engineering

- CGPA: 7.5 / 10

PROFESSIONAL DEVELOPMENT

Actively expanding into React.js, Azure Cloud, Docker, Microservices Architecture, and CI/CD pipelines to broaden modern full-stack and cloud-native delivery capability.

LANGUAGES

- Tamil (Native / Bilingual Proficiency)
- English (Professional Working Proficiency)